BUILDING WITH AI

Use AI to build the machine.
Not to *be* the machine.

Mainframes, pacemakers and car ECUs ran for generations without a single token.

That's my bar: AI helps me make it. Nothing calls a model when you use it.

swipe →

LESSON 1

Spend tokens once, at build time.

A model call at run time is a bill, a latency, and an outage on every click, forever.

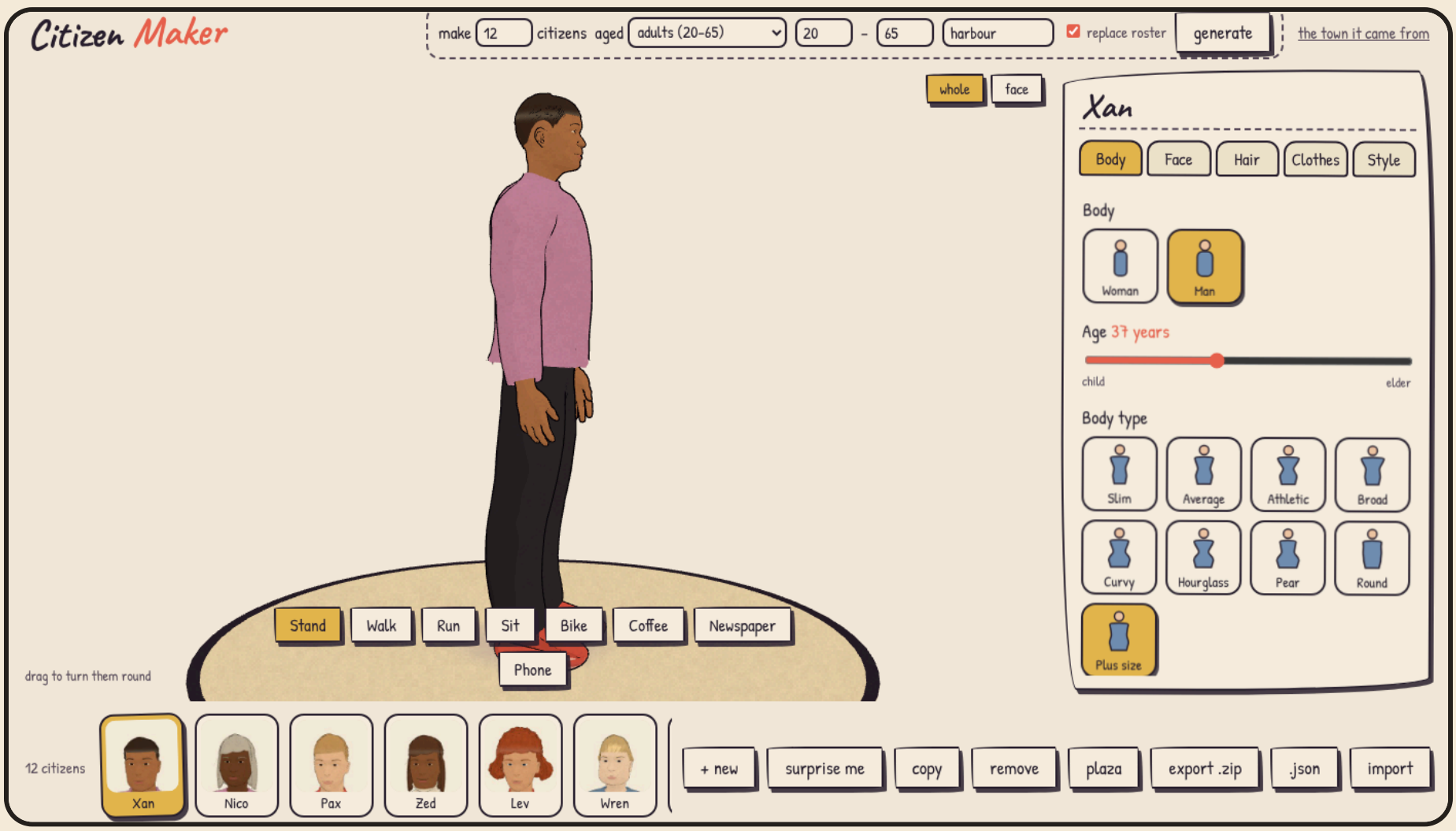
A model call at build time is paid once. What it leaves behind is code that runs for free.

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model calls when anyone uses the projects in this deck

EXAMPLE · CITIZEN MAKER

Game characters on demand, no AI inside.



Up to 200 rigged, animated people per batch, 10 drawing styles, exported as .glb. Plain three.js on GitHub Pages: no server, no API key. Built with an agent; runs on math.

LESSON 2

Replace "generate" with "seed".

Ask an LLM for 30 villagers and you get 30 new villagers each time, and a new bill.

Citizen Maker takes a seed: `?batch=30&seed=harbour` makes the same crowd, every time, on any machine.

Reproducible means testable. Testable means shippable.

EXAMPLE · EQUALIZER

Beats from math.



No samples, no generative audio. Every drum and melody is an equation you can read and edit, computed 44,100 times a second. AI helped write the synth; the synth needs no AI.

LESSON 3

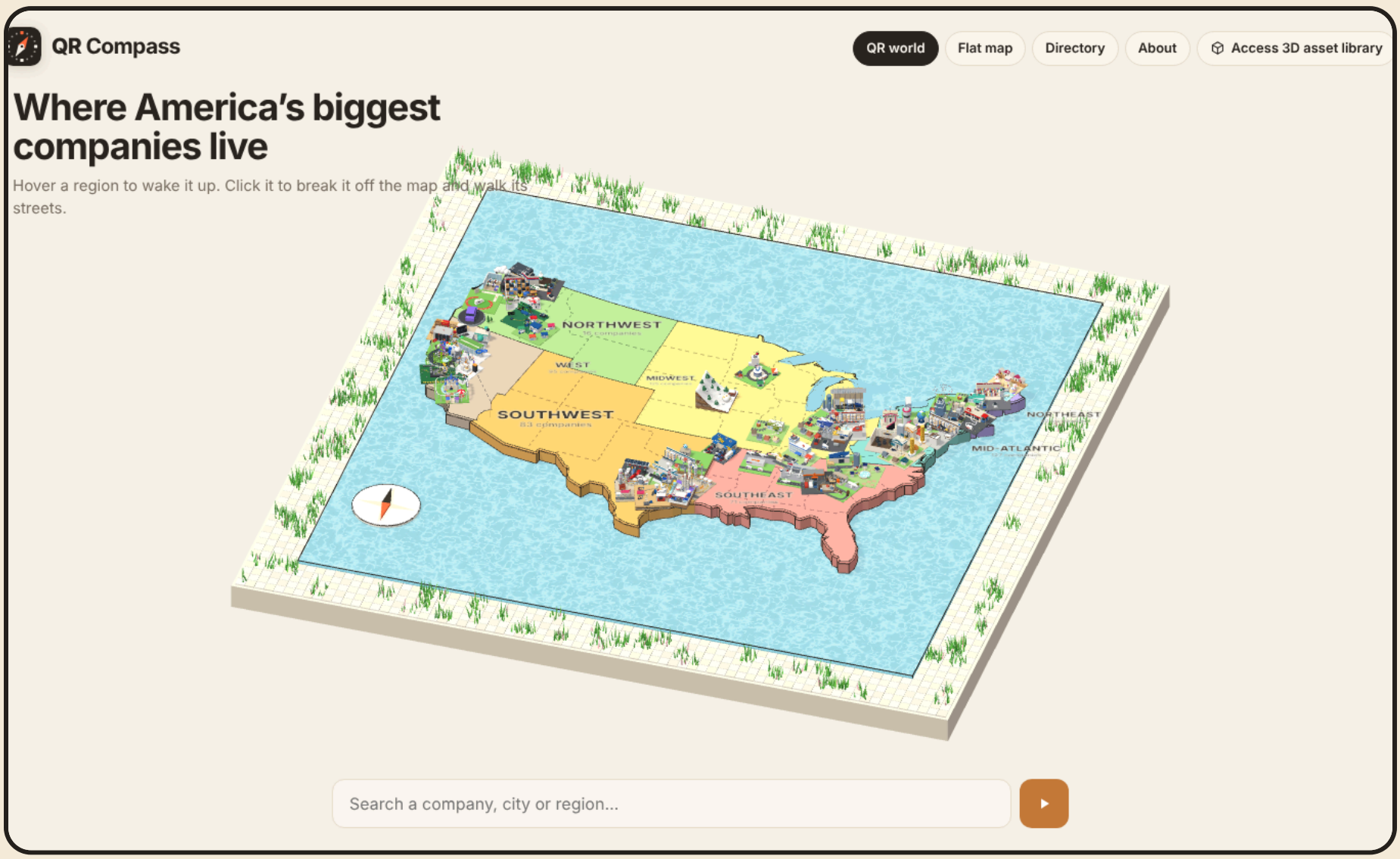
Ask the agent for tools, not answers.

- Not "describe this company's logo" ×500 — a `build.mjs` that writes all 500 pages (QR Compass).
- Not "tell me which assets behave alike" — a `run_all.sh` that regenerates every number on a CPU (Behaviors).
- Not "load the body data" — a bake script that packed it 3.9 MB → 1.85 MB, lossless.

An answer is used once. A tool is used every day.

EXAMPLE · QR COMPASS

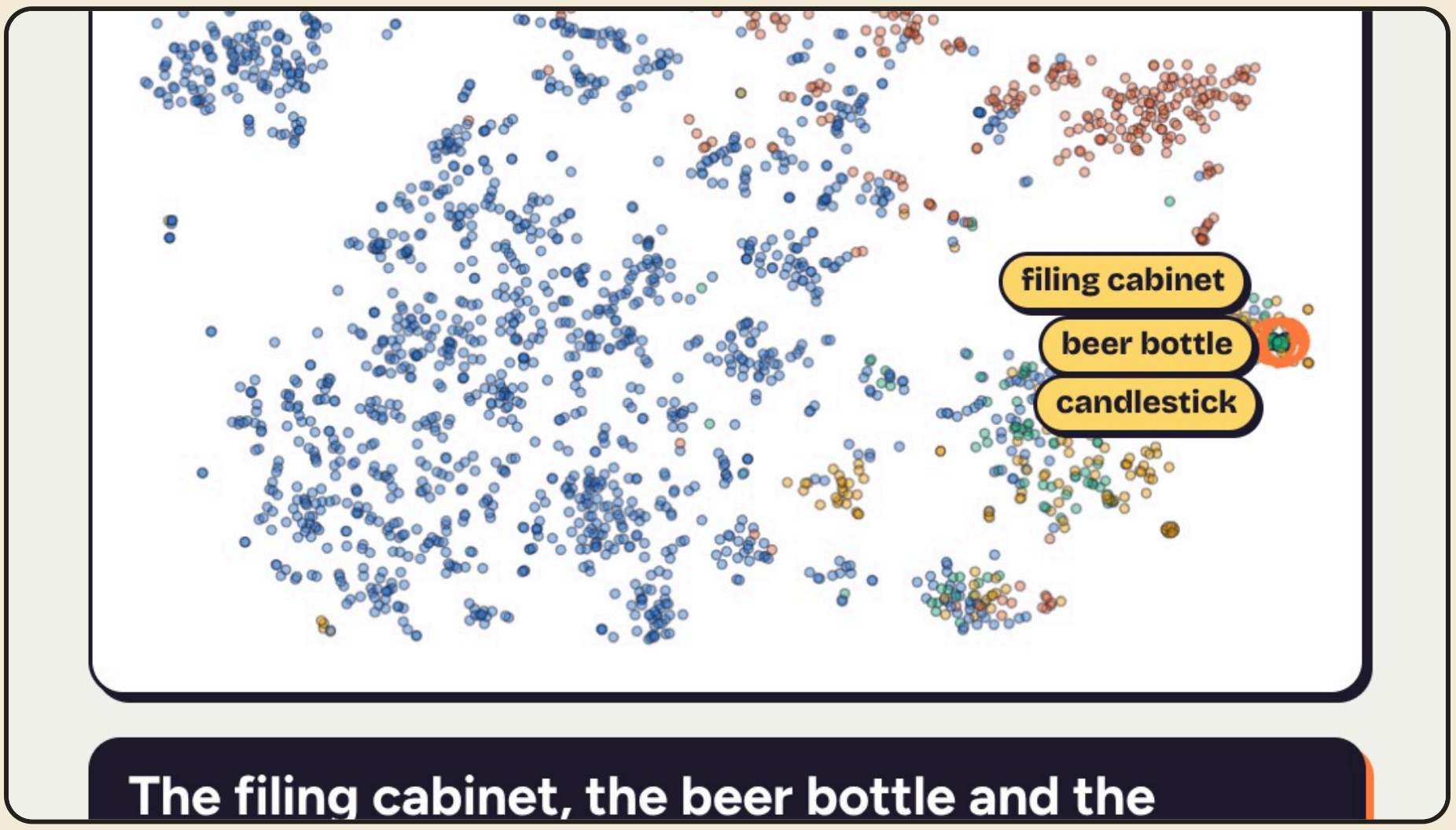
The Fortune 500, as a static site.



Every company gets a 3D model that turns into its own scannable QR code. All generated at build time from open data, then served as plain files.

EXAMPLE · BEHAVIORS

Search 2,166 3D assets by how they move.



Simulated pushes, ramps and drops, summarized into a searchable index. The learning happens offline on a CPU; the live demo is a lookup.

LESSON 4

Write the rules down, so you stop paying to re-explain them.

Every repo gets an `AGENTS.md`: how to deploy, what to check before pushing, what never to do.

"Pivot limbs at the joint." "Check all four compass angles." "Bump the cache token."

Short context, fewer retries, fewer tokens per session.

LESSON 5

Hand off to files, not to a chatbot.

Citizen Maker's export zip carries a `HANDOFF.md`, a manifest and a viewer, so the next person (or agent) can drop the models into Unity, Unreal, Godot or Blender without asking anyone.

The output outlives the conversation that made it.

THE CHECKLIST

Before you put a model in the loop, ask:

- Could this be a lookup, a formula or a seed?
- Could the AI write the code once instead of answering every time?
- Will it still work with the API key revoked?
- Can I reproduce yesterday's output exactly?

AI is a phenomenal builder. Most products just need what it builds.

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